

Problem Statement

Build a comprehensive AI interview-prep platform with 20 advanced features across 5 categories: Cognitive & Psychological tools, Objective Assessment Engines, Simulation & Environment, Advanced Tech & Integration, Long-term Growth & Community. Multilingual, animated, fully interactive.

Architecture

- **Backend:** FastAPI + MongoDB + emergentintegrations (Claude Sonnet 4.5)
- **Frontend:** React 19 + Tailwind + framer-motion + Recharts + Phosphor Icons + react-router 7
- **AI:** Claude Sonnet 4.5 via Emergent LLM key
- **Voice:** Browser Web Speech API (SpeechRecognition)
- **Webcam:** getUserMedia for micro-expression panel
- **Auth:** None (local profile, fastest MVP)
- **i18n:** 5 languages — EN/ES/FR/HI/DE — custom context

User Persona

Pre-interview candidates (students, job-switchers, bootcamp grads) preparing for FAANG-style or competitive interviews.

Core Requirements

- 20 modular features implemented as either fully functional flows or interactive demos
- Real LLM-powered scoring (STAR, fluff, confidence, rewrite suggestion)
- Dark "flight-simulator / neuroscience lab" aesthetic — Outfit + IBM Plex Sans + JetBrains Mono
- Multilingual support (5 languages)
- Animated micro-interactions (framer-motion staggered entrances)

What's Been Implemented (Iteration 1 — 2026-02)

Backend (13 endpoints, 100% tested)

- `/api/interview/start` — LLM generates 5 tailored questions per persona
- `/api/interview/answer` — STAR scoring, fluff/wobble/pause analysis, confidence, AI rewrite
- `/api/jd/scrape` — URL scrape + LLM extracts role/stack/sample questions
- `/api/resume/check` — Consistency vs verbal transcript
- `/api/reverse/score` — Scores the questions candidate asks the interviewer
- `/api/pitch/score` — 30-second elevator pitch scoring
- `/api/imposter/score` — Perceived vs actual ability gaps
- `/api/glassdoor/{company}` — Curated real-talk tips
- `/api/readiness/calculate` — Aggregate readiness passport score
- `/api/mentor/match` — AI mentor pairing
- `/api/profile` — Save/load local profile
- `/api/shadow` — "Perfect" candidate playback

Frontend

- **Landing** (`/`) — Hero, live HUD card, 6 feature lenses, animated stats
- **Dashboard** (`/dashboard`) — Readiness passport, knowledge radar (Recharts), HRV trend area chart, feature tiles
- **Live Interview** (`/interview`) — Camera feed, live biometric HRV panel, wobble alerts, fluff highlighting, STAR score panel, persona switcher with curveball mid-interview switch animation
- **Modules** (`/modules`) — 11 interactive tool cards (JD Scraper, Resume Check, Reverse Interviewer, Pitch Gauntlet, Imposter Meter, Shadow Interviewer, Mock Battles, Blind Technical, Mentor Matching, Glassdoor Tips, VR Selector)
- **Passport** (`/passport`) — Verified shareable readiness passport card

Backlog (P1/P2 — future)

- **P1:** Persistent user accounts (JWT or Emergent Google Auth)
- **P1:** Mock battle PvP real-time WebSocket sync
- **P1:** Real Glassdoor scraping (currently curated)
- **P1:** Real Apple Watch / Fitbit OAuth integration (currently camera/simulator)
- **P2:** True WebGL VR office (currently styled selector)
- **P2:** Whisper STT fallback when browser Web Speech unavailable
- **P2:** Rate limiting on LLM endpoints
- **P2:** PDF export of readiness passport

Next Actions

1. (Optional) Add JWT auth + persistent score history per user
2. (Optional) Real wearable integration via Apple HealthKit / Fitbit OAuth
3. (Optional) WebSocket-based PvP Mock Battles "